Dynamic memristor for bioinspired sensory memory

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Heading toward a future society reliant on large-scale data processing, existing computing architectures face limitations due to the disparity in processing speeds between memory and the central processing unit (CPU), commonly referred to as the von Neumann bottleneck. Consequently, researchers have explored alternative computing architectures, with neuromorphic computing emerging as a promising, power-efficient solution. However, to further enhance data processing efficiency and expand memory capacity, additional studies are imperative. Sensory memory, a type of memory that briefly retains sensory information following the cessation of the original stimulus, plays a crucial role. Serving as a temporary buffer, it stores sensory impressions for a fraction of a millisecond to facilitate subsequent processing. Due to its temporary nature, sensory memory exhibits significant capacity while also enabling rapid synaptic plasticity. To leverage this efficiency, the TiN/WO_x/FTO memristor demonstrated an average HRS/LRS memory window of 14.51 during a 100-cycle endurance test, without significant resistance fluctuations. Throughout consistent resistive switching, a memory window greater than 10 was consistently maintained with minimal fluctuations in resistance states, as indicated by the coefficient of variation values, thereby demonstrating the uniformity of the TiN/WO_x/FTO memristor under continuous external bias application. Furthermore, cell-to-cell uniformity was confirmed by measuring 12 random cells over 30 cycles. Moreover, we employed our TiN/WO_x/FTO device, known for its rapid facilitation, to investigate various synaptic behaviors. Our study examined the impact of spike arrival on synaptic weight changes under different pulse conditions, including amplitude, width, and interval. Additionally, we explored high-density memory adaptation through a 14×14 synapse image. Finally, we employed a four-state reservoir computing pattern recognition system to assess the computing capabilities and multifunctional behaviors of our memristor. In conclusion, our findings suggest that the TiN/WO_x/FTO memristor holds promise as a volatile memory device capable of adapting sensory memory for use in neuromorphic computing applications.

Impact statement

Recent advancements in resistive random-access memory devices have led to their application in neuromorphic computing, leveraging their synapse-emulating and energy-efficient properties. Some studies have highlighted the sensory memory applications of memristor devices. By incorporating biological functions, memristors have become instrumental in neuromorphic and visual memory applications. To further enhance data processing efficiency and expand memory capacity, additional studies are imperative. Sensory memory, a type of memory that briefly retains sensory information following the cessation of the original stimulus, plays a crucial role. Serving as a temporary buffer, it stores sensory impressions for a fraction of a millisecond to facilitate subsequent processing. Due to its temporary nature, sensory memory exhibits significant capacity while also enabling rapid synaptic plasticity. To leverage this efficiency, we employed our TiN/WO_x/FTO device, known for its rapid facilitation, to investigate various synaptic behaviors. Our study examined the impact of spike arrival on synaptic weight changes under different pulse conditions, including amplitude, width, and interval. Additionally, we explored high-density memory adaptation through a 14×14 synapse array. Finally, we employed a four-state reservoir computing pattern recognition system to assess the computing capabilities and multifunctional behaviors of our memristor. In conclusion, our findings suggest that the TiN/WO_x/FTO memristor holds promise as a volatile memory device capable of adapting sensory memory for use in neuromorphic computing applications.

Introduction

As technology advances, the volume of temporal data processed every second continues to grow, rendering the current computing architecture, which separates the CPU and memory unit, obsolete. To facilitate cost-effective data processing, scientists are striving to replicate the data processing functions of the human brain.[1,2] In the biological brain, interconnected neurons and synapses enable rapid parallel processing of incoming data from the sensory units of our body. For such neuromorphic computing applications, resistive random-access

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memory (ReRAM) stands out among other next-generation memories due to its CMOS compatibility, fast switching speed, high endurance, long retention time, and simple two-terminal structure.[3–9] Memristors are primarily used to emulate brain functions by testing linear and symmetric synaptic weight updates and implementing the memory properties observed in the biological brain. Upon the arrival of incoming spikes in interconnected synapses, synaptic plasticity changes, allowing for the categorization of incoming information into sensory, short-term, and long-term memory.[10,11] Sensory memory, a crucial component in this emulation, briefly retains sensory information from our environment after the original stimulus has ceased. It acts as a temporary buffer, holding sensory impressions for a brief period to facilitate further processing.[12,13] This type of memory boasts a large capacity but has a short duration, typically lasting only milliseconds to several seconds. Sensory memory plays a vital role in perception and cognition by continuously supplying our brain with information, serving as the foundation for higher cognitive functions such as attention, perception, and memory encoding.[14] In contrast, short-term memory has a longer duration compared to sensory memory and transitions depending on the attention given to sensory stimuli. Typically lasting from milliseconds to minutes, short-term memory allows synapses to perform important functions such as rapid response and information filtering.[10] The transition from short- to long-term memory occurs when the duration and intensity of stimuli reach threshold values, often through a rehearsal process. Long-term memory can last for days, shaping our experiences throughout our lives. Despite frequent studies on short- and long-term memory, there has been little focus on utilizing sensory memory.[15–18] Given its large capacity, leveraging sensory memory could offer advantages for effective neuromorphic computing.

In RRAM, data storage relies on the resistive switching phenomenon occurring within the memristor, typically in the insulating layer positioned between the top and bottom electrodes. One commonly used insulating film for resistive switching is metal oxide, where the process involves the migration of oxygen ions within the metal oxide films, leading to variations between two resistance states: the high-resistance state (HRS) and the low-resistance state (LRS).[19–21] In metal oxides, the resistive switching phenomenon can vary depending on the presence of a conductive filament. In filament-based RRAM, oxygen ions migrate under bias application, leaving behind oxygen vacancies due to redox reactions. These vacancies accumulate and form a conductive pathway connecting the top and bottom electrodes, which breaks under opposite bias application.[22] Conversely, nonfilamentary switching RRAM exhibits changes in conductance state through the increase and decrease of defective regions formed under bias application based on oxygen ions.[23] However, filamentary-based RRAM faces significant challenges, primarily due to the inherent randomness and abruptness within the conductive filament.[24] Filament-based RRAMs encounter difficulties in achieving linear and symmetric updates of synaptic weight due to the abrupt rupture and formation of filaments. In contrast, interface-type RRAM devices typically exhibit gradual resistive switching, which facilitates easier modulation of synaptic weight, making them more suitable for synapse emulation.[25] Metal oxides such as TiO_x, TaO_x, and WO_x are commonly used for interface-type switching.[26–28] For instance, Kim et al.[29] demonstrated a sodium-doped titania interface-type memristor with high self-rectifying properties. Sahu et al.[30] reported on an Ag/TiO_2/Pt memristor operating based on Ag ion migration, which exhibited synaptic functions such as spike-time-dependent plasticity with nociceptor emulation. Zhao et al.[31] presented an AlO_x/TaO_x self-rectifying memristor capable of linear and symmetric updates of conductance. Lin et al.[32] developed a flexible Pt/WO_x/Ti memristor demonstrating learning and forgetting functions similar to those of the biological brain, including paired-pulse facilitation (PPF). Additionally, Kim et al.[33] introduced a Ni/WO_x/ITO volatile memristor suitable for adapting reservoir computing.

In this study, we engineered a memristor capable of emulating synapses using TiN/WO_x/FTO stacks on a glass substrate. Prior to simulating synapses, we conducted tests on uniformity, a crucial parameter for memory devices, by applying DC bias to ensure consistency across cycle-to-cycle and cell-to-cell operations. Leveraging the rapid facilitation of our TiN/WO_x/FTO device, renowned for its efficiency, we explored various synaptic behaviors by examining how spike arrival affects synaptic weight changes under different pulse conditions. By adjusting the amplitude, width, and interval of the spike, we implemented spike-rate-dependent plasticity (SRDP), spike-width-dependent plasticity (SWDP), and spike-amplitude-dependent plasticity (SADP), thereby achieving various synapse emulations. Furthermore, leveraging the volatile nature of the memristor, we demonstrated learning and forgetting functions to emulate the biological brain. Additionally, we investigated the adaptation of high-density memory through a 14×14 synapse image and evaluated the computational capabilities and multifunctional behaviors of our memristor using a four-bit reservoir computing pattern recognition system. Our findings indicate that the TiN/WO_x/FTO memristor shows promise as a volatile memory device suitable for integrating sensory memory into neuromorphic computing applications.

Experiments

Via the subsequent fabrication process, the TiN/WO_x/FTO memristor was formed. Initially, a commercially available 2.2-mm-thick fluorine-doped tin oxide (FTO) bottom electrode on a glass substrate was prepared and cleaned using isopropyl alcohol (IPA) and acetone. Subsequently, a layer of the insulating film WO_x was deposited onto the FTO electrode using a DC reactive sputtering process under a main chamber pressure of 3 mTorr, utilizing a W target with 99.99% purity. The sputter power was set to 80 W, with Ar flow at 20 sccm and O at 15 sccm. Next, the top electrode

TiN was patterned using negative PR and photolithography techniques, followed by deposition and liftoff processing in acetone. For TiN deposition, an RF reactive sputtering process was employed with a Ti target purity of 99.99%, under a main chamber pressure of 3 mTorr, RF power of 100 W, and Ar (19 sccm) and N_2 (1 sccm) gases. During the experimentation, voltage bias was applied to the top TiN electrode, while the bottom FTO electrode was grounded, using a semiconductor parameter analyzer (Keithley 4200-SCS and 4225-PMU ultrafast module, Solon, Ohio), conducted at room temperature and pressure. For x-ray photoelectron spectroscopy (XPS) depth analysis, a Nexsa instrument equipped with a Microfocus monochromatic x-ray source (Al-K [1486.6 eV]), a sputter source (Ar^+), an ion energy of 2 kV, and a beam size of 50 m was utilized. Additionally, a cross-sectional transmission electron microscopy (TEM) image was obtained to examine the structural characteristics of the fabricated memristor (Oxford Instruments, Tubney Woods, Abingdon, UK).

Results and discussion

In **Figure** 1a, the schematic depicts the TiN/WO_x/FTO memristor fabricated in sequence on a transparent glass substrate. Figure 1b displays a cross-sectional TEM image illustrating the TiN, WO_x, and FTO layers deposited sequentially, with each TiN and WO_x layer measuring 100 and 60 nm, respectively. The investigation further delves into the chemical composition of the insulating WO_x film, wherein the migration of oxygen ions induces resistive switching under external bias. Figure 1c–d presents XPS spectra of W 4f and O 1s, confirming the presence of the WO_x layer. Notably, the O 1s spectra in Figure 1d reveal double peaks at binding energies around 530.36 and 530.76 eV, indicating metal–oxygen (M–O) bonding and the presence of oxygen vacancies (V_o^+) within the insulating film.[34] These findings validate the existence of a defective WO_x layer, facilitating the memory functionality of the TiN/WO_x/FTO memristor. Additionally, Supplementary information (SI) Figure S1 showcases O 1s spectra obtained via depth mode, depicting different etch times.

Figure 2a illustrates the electrical characteristics of the TiN/WO_x/FTO memristor under DC bias conditions, displaying typical bipolar I–V curves. When a positive bias (2.5 V) is applied to the top electrode, a gradual increase in current occurs, indicating a resistive switching phenomenon from HRS to LRS. This contrasts with the abrupt and rapid current increase observed in filament-based RRAM devices, enabling gradual and easy modulation of resistance states.[24] Upon reaching the compliance current (CC) of 1 mA, which was utilized to avoid the hard breakdown of the memristor, the current gradually returns to a higher level, signifying the occurrence of resistive switching, known as the "set" process. Conversely, applying a negative bias (−4 V) results in a gradual decrease in current, creating a significant window. This indicates the transition of the memristor's resistance state from LRS to HRS, thereby limiting current flow, referred to as the "reset" process. By repeating the set and reset processes, the uniformity of the device's resistance state during consistent resistive switching phenomena can be observed. In this investigation, the uniformity of consistent switching properties is identified as the endurance characteristic of the memristor, evaluated through a read bias of −0.7 V following each set and reset process under DC bias application of 2.5 and −4 V. Figure 2b illustrates the endurance performance of the TiN/WO_x/FTO memristor, with an average window between HRS and LRS of 14.51. Moreover, the variability in endurance is quantified by the coefficient of variation (CV), depicted in Figure 2c, showing CV values of 0.11 and 0.19 for LRS and HRS, respectively. Throughout consistent resistive switching, a memory window of over 10 is consistently maintained with minimal fluctuation in resistance states, as indicated by the CV values, demonstrating the uniformity of the TiN/WO_x/FTO memristor under continuous external bias application. Additionally, in SI Figure S2, the cell-to-cell uniformity of the TiN/WO_x/FTO memristor is examined by applying the same set and reset bias to randomly selected 12 cells, each subjected to 30 DC cycles. The endurance performance across different cells, evaluated from a read bias of −0.7 V, is illustrated in SI Figure S3, revealing similar resistance state values between HRS and LRS across different cells and repeated cycles. Interestingly, in the retention test conducted by applying a consistent read bias (−0.7 V) after the set process, we observed a gradual decrease in the LRS current (Figure 2d). This observation indicates the volatile characteristic of our TiN/WO_x/FTO memristor, with a retention loss of 99% over 1300 s. The term retention loss was acquired through the following equation:

$$\text{Retention loss}(\%) = \left(\frac{I_{1300} - I_0}{I_0}\right) \times 100, \qquad 1$$

where terms I_{1300} and I_0 denote the current values acquired after 1300 s and at the beginning of the retention experiment, respectively. The underlying conduction mechanism driving the observed resistive switching properties of the TiN/WO_x/FTO memristor can be elucidated by the migration of oxygen ions and the expansion of defective regions, rendering the device conductive. In contrast to filamentary-based resistive switching devices, which exhibit abrupt current increases with fluctuations in the I–V curve due to the stochastic nature of filament formation and rupture,[24,35] it is plausible to interpret that the TiN/WO_x/FTO memristor operates through a non-filamentary switching mechanism, predominantly known as interface-type switching.[25] In interface-type switching, the application of bias to the memristor's electrode induces the migration of oxygen ions within the metal oxide films under the influence of the electric field, leading to the formation of defective regions within the thin film and subsequently increasing the device's conductance. Conversely, under the application of opposite bias, the oxygen ions are repelled back to their original positions. Additionally, in interface-type memristors, upon the removal of external bias, self-diffusion of ions

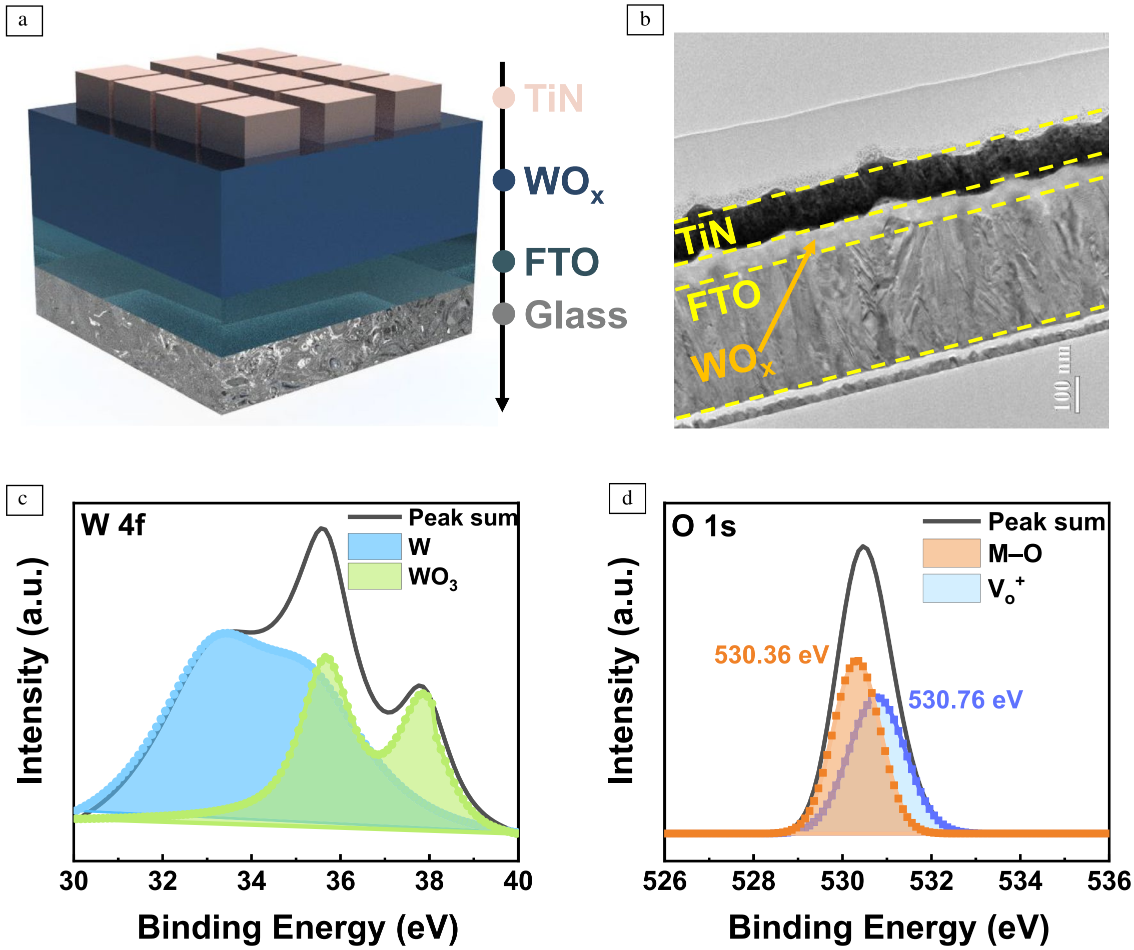

Figure 1. (a) Schematic illustration of the TiN/WO_x/FTO memristor. (b) Cross-sectional transmission electron microscopy image of the TiN/WO_x/FTO memristor. X-ray photoelectron spectroscopy spectra of the WO_x thin film. (c) W 4f and (d) O 1s.

occurs due to the concentration gradient, resulting in a gradual return to their original positions and consequently causing the memristor to gradually insulate.[36] Similar to the reported mechanism, in the device's initial state (SI Figure S4a), oxygen ions are drawn toward the top electrode when a positive bias is applied to the TiN electrode, potentially migrating toward the TiON layer. This layer is typically formed due to the high oxygen scavenging properties of the TiN electrode.[37] During consistent resistive switching, the TiON layer serves as a reservoir for oxygen, storing and releasing oxygen ions. This process contributes to the uniformity during repeated cycles.[38,39] As a result of oxygen ion migration, the expansion of the defective region occurs, rendering the device conductive and reducing its resistance state, thereby transitioning the memristor into the LRS (SI Figure S4b). Conversely, when a negative bias is applied, the migrated oxygen ions are repelled, reducing the defective region, and insulating the device, leading to the memristor transitioning into the HRS (SI Figure S4c).

Therefore, the expansion and contraction of the oxygen defect region on the FTO side occur in response to the migration of oxygen ions. Moreover, the volatile nature of the TiN/WO_x/FTO memristor can be elucidated by the self-diffusion of oxygen ions due to the concentration gradient, gradually returning to their original state, thus gradually transitioning from LRS to HRS. Leveraging this volatile behavior, a consistent set process with a delay (relaxation) time can be implemented. Figure 2e illustrates 10 sequential set processes using a relaxation time of 10 s. Although the 10-s relaxation time may not have been sufficient for the full return of oxygen ions, resulting in a partial increase in HRS values over repeated cycles, the successful experimentation of 10 sequential I–V curves utilizing relaxation is still observable. Moreover, by controlling the CC during bias application to the top electrode, it is possible to observe multilevel cell (MLC) characteristics, which result from different rates of oxygen ion migration. When the CC is high, a large amount of oxygen ions could migrate, creating a significant defective region and increasing

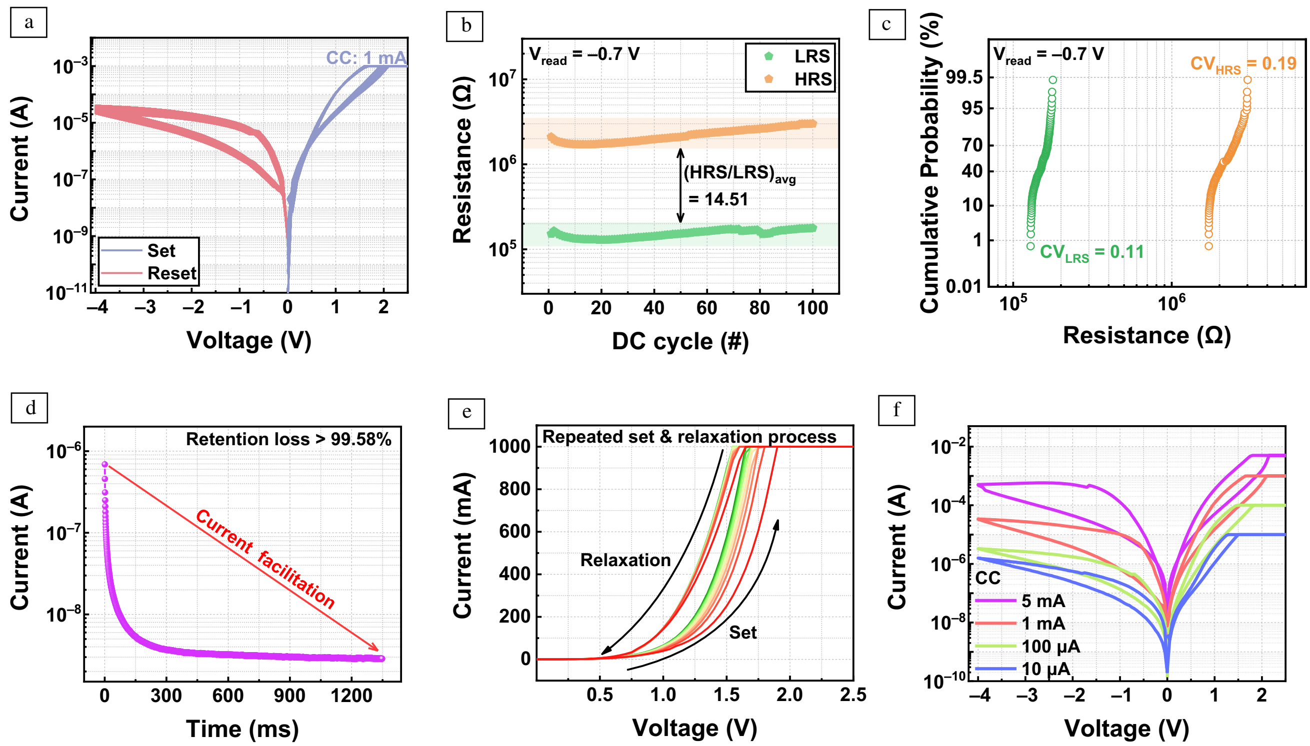

Figure 2. (a) I–V curve of the TiN/WO_x/FTO memristor, acquired through DC measurement. (b) Endurance properties of the memristor observed for 100 DC cycles, gained from $V_{read} = -0.7$ V. (c) Cumulative probability of high-resistance state (HRS) and low-resistance state (LRS) during consistent resistive switching. (d) Retention properties of the TiN/WO_x/FTO memristor, showcasing current facilitation. (e) Sequential set process utilizing relaxation. (f) Multilevel cell properties of the TiN/WO_x/FTO memristor, acquired through differing compliance current (CC).

the LRS current. Conversely, a low CC limits the migrated oxygen ions, resulting in a smaller defective region and a smaller LRS current. Figure 2f illustrates the MLC properties of the TiN/WO_x/FTO memristor, obtained by varying the CC under four different conditions, highlighting the memristor's adaptability for achieving large storage density.

To evaluate the capability of implementing artificial synapses using the developed memristor, various synapse-like functionalities of the RRAM device were realized through electrical pulse training. The constructed pulse train is presented in SI Figure S5a. Following this train, 50 potentiation pulses and 50 depression pulses were applied. Figure S5b illustrates typical potentiation and depression curves, achieved through sequential pulses to obtain gradual conductance changes, emulating the synaptic weight update process in the biological brain.

Considering the volatile characteristics of our TiN/WO_x/FTO memristor, it could be possible to emulate the memory functions of our brain, where different reactions occur depending on the input stimuli. For example, as depicted in **Figure 3**a, sensory inputs received through the sensory units of our body (mouth, nose, eyes) are transmitted to our brain through changes in synapse weights in interconnected neurons and synapses, resulting in sensory memory formation. Sensory memory has a large capacity but is easily forgotten, leading to synaptic weight facilitation lasting about milliseconds.[11,40,41] When attention is directed toward sensory memory, the transition from sensory memory to short-term memory takes place, lasting from milliseconds to several minutes before gradually fading away over time.[10] One of the most fundamental behaviors of the volatile memristor utilized for memory that is forgotten is PPF. The phenomenon known as PPF exhibits a resemblance to the correlation observed between temporal spike pairs.[10] In the emulation of memristor behavior, this process is interpreted by applying twin pairs of pulses at different intervals and assessing the degree of enhancement. The twin pulse, having an amplitude of 1.5 V, was applied for 1 ms, with the interval between pulses ranging from 1 μs to 5 ms. To explain this phenomenon, the term "PPF index" was employed, calculated using the following equation:

$$\text{PPF index}(\%) = \left(\frac{I_2 - I_1}{I_1}\right) \times 100. \qquad 2$$

The terms I_2 and I_1 represent the current response following the application of the second and first pulses, respectively, indicating the relative enhancement of the second pulse compared to the first. The PPF index, obtained as a function of spike interval, is depicted in Figure 3b, shows that longer

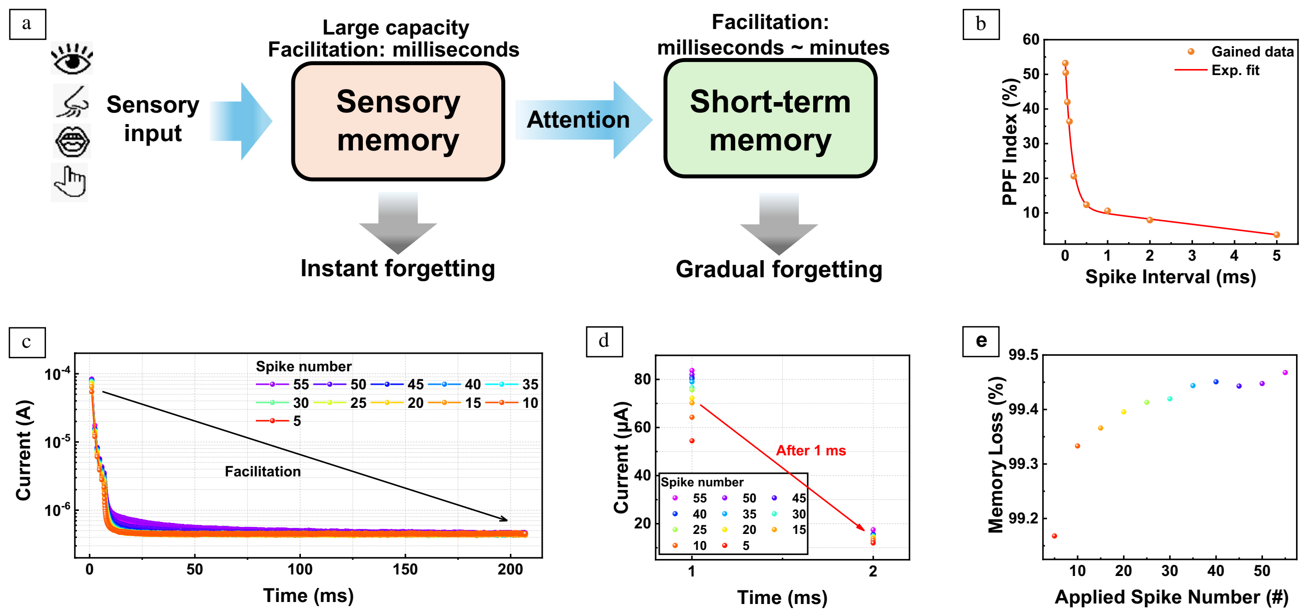

Figure 3. (a) Schematic illustration of the memory categories, representing sensory memory and short-term memory. (b) Paired-pulse facilitation (PPF) acquired as a function of spike interval. (c) Sensory memory functions of the TiN/WO_x/FTO memristor. (d) Facilitation of current after 1 ms. (e) Memory loss as a function of applied spike number, acquired after 100 ms.

intervals result in the forgetting of the previous pulse application, leading to a gradual decrease in the rate of enhancement (PPF index). Next, the volatile properties of the TiN/WO_x/FTO memristor were investigated under pulse applications. Figure 3c demonstrates the current facilitation obtained by applying a pulse scheme consisting of varied set pulse (−2 V, 1 ms) numbers followed by sequential read pulses (−0.7 V) for 220 ms. Interestingly, as illustrated in Figure 3d, although the response read current immediately after the set pulse application varies depending on the number of spikes, significant facilitation occurs within 1 ms, resulting in a substantial current facilitation. Furthermore, Figure 3e presents the memory loss obtained through the retention loss equation in Equation 1, where all applications under different spike numbers result in memory loss of over 99% after 100 ms. This underscores the rapid forgetting speed of the TiN/WO_x/FTO memristor and its suitability for emulating the functions of sensory memory, capable of rapid forgetting while maintaining large storage capacity.

Utilizing the sensory memory properties of our memristor, we emulated the learning and forgetting behaviors observed in the human brain through sequential training and sequential forgetting. As depicted in **Figure 4**a, we sequentially applied 10 training pulses (1.7 V, 1 ms) and forgetting pulses (0 V, 100 μs) to the memristor, with a read pulse of −0.7 V following each sequence to record changes in current. Despite the rapid forgetting of sensory memory, resulting in minimal changes during each forgetting scheme, a slight synaptic plasticity remained after each learning and forgetting cycle, leading to some enhancement in learning behavior and an increase in current response after each learning sequence. Moreover, in the TiN/WO_x/FTO memristor, easy modulation of such synaptic weight changes is achievable due to its gradual resistive switching properties, allowing for controlled synaptic weight updates under various learning conditions, as illustrated in Figure 4b–c. In these figures, the learning and forgetting behavior depicted in Figure 4a was varied by altering the learning pulse width across six different conditions, ranging from 500 to 5 ms. The results effectively mirror the functions of our brain, wherein longer learning pulses lead to larger synaptic weight changes that persist for a longer duration. The term "synaptic weight change" is defined by the following equation:

$$\text{Synaptic weight change}(\%) = \frac{I_f}{I_i} \times 100. \quad (3)$$

The terms I_f and I_i represent the current after and before learning, respectively. Additionally, in the context of adapting neuromorphic computing, it is crucial to replicate the functions of synapse actions within the brain, as the foundation lies in mimicking the efficient data processing capabilities of the biological brain. To achieve this goal, various synapse emulation techniques were tested, each differing in spike interval, spike width, and spike amplitude. These techniques include SRDP, SWDP, and SADP. Firstly, SRDP simulates synaptic weight changes influenced by the spike interval. For this investigation, 10 sequential pulse patterns of 1.7 V and 1 ms were employed, with intervals spanning seven different conditions

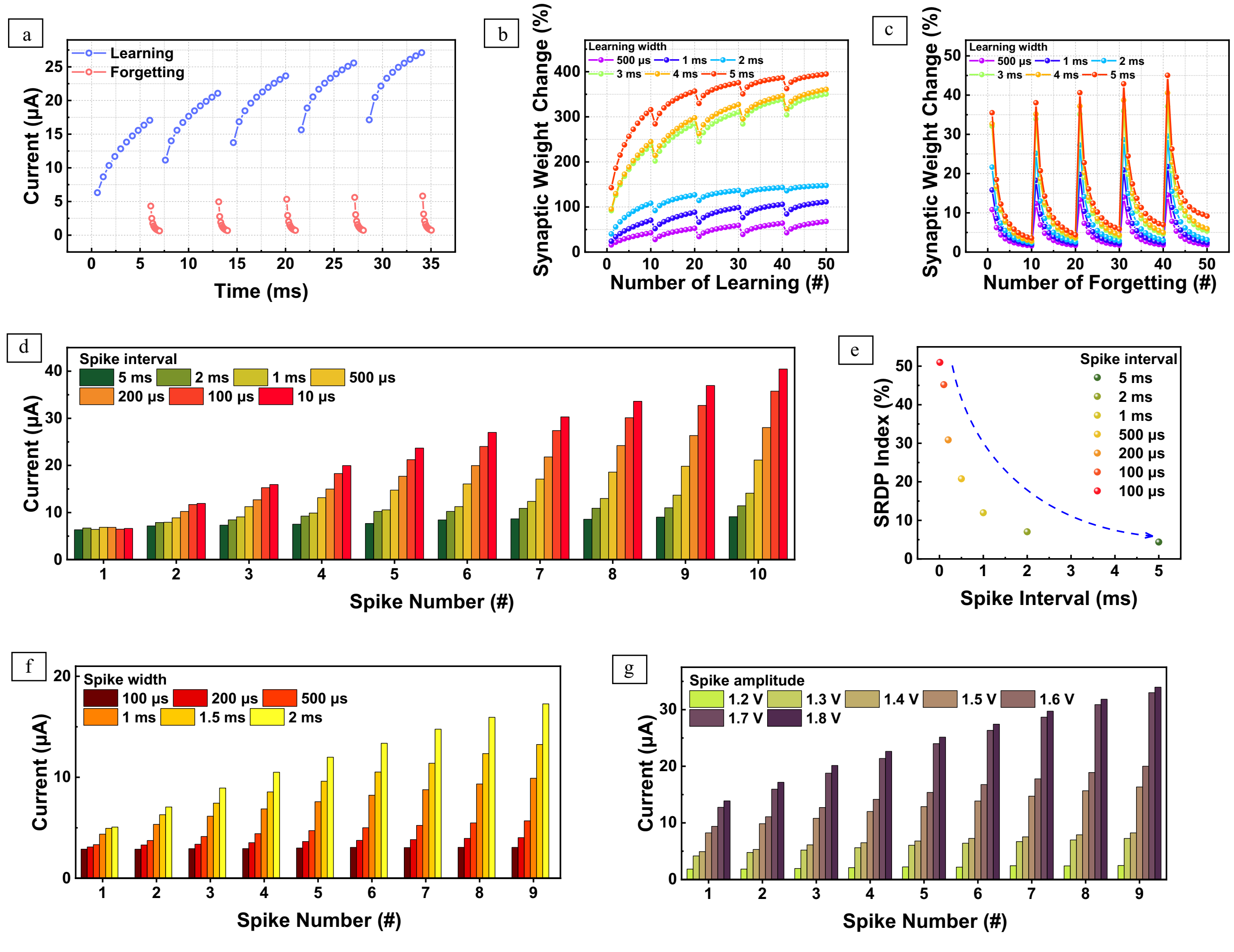

Figure 4. (a) Learning and forgetting process represented through the TiN/WO_x/FTO memristor. (b) Comparison of synaptic weight increase depending on the learning width. (c) Comparison of synaptic weight decrease depending on the learning width. (d) Spike-rate-dependent plasticity (SRDP) function of the TiN/WO_x/FTO memristor. (e) SRDP index represented as a function of spike interval. (f) Spike-width-dependent plasticity function of the TiN/WO_x/FTO memristor. (g) Spike-amplitude-dependent plasticity function of the TiN/WO_x/FTO memristor.

ranging from 10 to 5 ms. The results in Figure 4d demonstrate that, akin to the findings in PPF, shorter intervals between spikes result in stronger enhancements, leading to larger current responses. The function of SRDP was quantified as the SRDP index in Figure 4e, calculated using the following equation:

$$\text{SRDP index (\%)} = \frac{A_{10} - A_1}{A_1} \times 100. \qquad 4$$

The terms A_{10} and A_1 represent the current after the 10th and 1st spike applications, respectively. It is noteworthy that there is a decrease in the SRDP index as a function of the spike interval, indicating that the enhancement rate over spikes strongly depends on the interval between the spikes. Moreover, moving on to Figure 4f–g, the impact of spike strength, characterized by spike width and amplitude, was investigated. In the case of SWDP (Figure 4f), the intervals between spikes and their amplitudes were fixed at 500 μs and 1.7 V, respectively, while varying the spike width across six conditions ranging from 100 μs to 2 ms. For SADP (Figure 4g), the intervals between spikes and their widths were fixed at 500 μs and 1 ms, respectively, while varying the spike amplitude. Across nine spike applications, both the SADP and SWDP functions demonstrated that the current response depends heavily on the strength of the incoming spike, highlighting the functionalities of biological synapses.[42]

The implementation of an image training using our TiN/WO_x/FTO memristor leverages the rapid facilitation and expansive memory capacity of sensory memory, as depicted in **Figure 5**. In Figure 5a, we present a 14 × 14 synapse image prepared for our memristor. For the training process and letter recognition, we employed the pulse scheme illustrated in Figure 5b. This involved programming with a set pulse of 2 V for 10 ms, followed by sequential read pulses of −0.7 V after

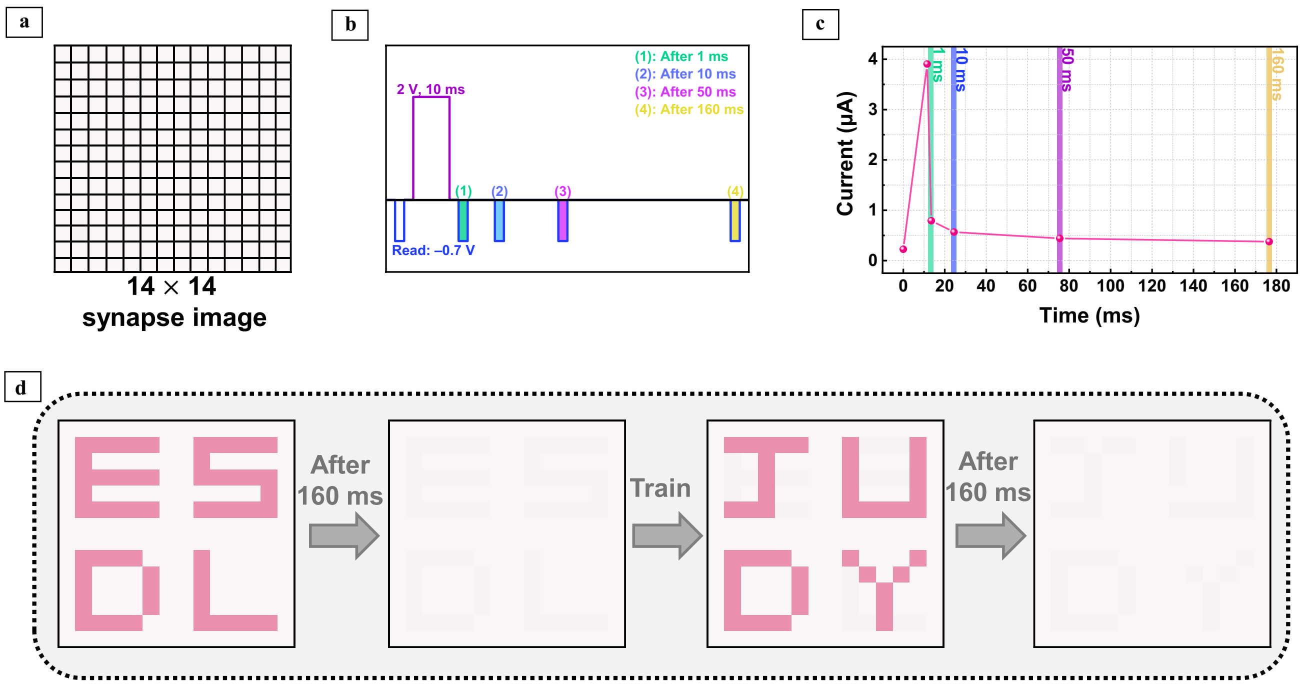

Figure 5. (a) Schematic illustration of the 14 × 14 synapse image implemented through TiN/WO_x/FTO memristor. (b) Illustration of the pulse scheme used for image training. (c) Current response of the applied pulse scheme in the TiN/WO_x/FTO memristor. (d) Training of different letters using the 14 × 14 synapse image.

relaxation periods of 1, 10, 50, and 160 ms, as demonstrated by the current responses in Figure 5c. During image training utilization, we represented only the current values obtained immediately after training and after 160 ms to efficiently demonstrate the sensory functions of our memristor, which enables rapid training and forgetting. The process of training the 14 × 14 synapse image is depicted in Figure 5d. Initially, the letters "E," "S," "D," and "L" were sequentially trained. However, due to the facilitation of trained current, these images tended to fade away after 160 ms of relaxation. Subsequently, the synapse image was reprogrammed by applying pulse signals, allowing for the programming of "J," "U," "D," and "Y," which also faded away after 160 ms of relaxation. Through the utilization of the synapse image, we effectively implemented high-density memory applications of the TiN/WO_x/FTO memristor, with repeatable training capabilities, harnessing its sensory memory properties.

Utilizing the nonlinear and sensory properties of the TiN/WO_x/FTO memristor, **Figure 6** illustrates the implementation of reservoir computing. Reservoir computing has emerged as a computing network to address challenges encountered by recurrent neural networks, such as training difficulties due to exploding or vanishing gradients in recurrent structures, as well as computational power and cost issues.[43] Employing a smaller neural network and a simpler weight update rule in the readout layer can help overcome the challenges faced by recurrent neural networks.[44–46] It consists of input, reservoir, and output layers, where input data undergo transformation in the reservoir layer through high-dimensional mapping. The reservoir layer, acting as the core component, preserves and transforms information from past time steps of input signals within a dynamic system network.[47] To implement reservoir computing with a memristor, two key priorities must be met. First, the memristor must exhibit nonlinear current response properties to enable weight updates within the reservoir layer through nonlinear dynamics. Second, because the input data transferred within the reservoir layer should be independent of past inputs, the memristor must possess volatile properties. In Figure 6a, when implementing reservoir computing with a memristor, pulse schemes comprising two distinct states, "0" and "1," are applied to the memristor to represent different reservoir states. These states are then conveyed through high-dimensional information in the TiN/WO_x/FTO memristor. The reservoir states of the TiN/WO_x/FTO memristor are obtained by applying "1" and "0" pulses of 1.7 and 0 V, respectively, lasting for 100 μs, with a subsequent read pulse of −0.7 V following each sequence to record the reservoir state. By varying the sequential application of "1" and "0" pulses, we obtained 16 distinct reservoir states, akin to four-bit reservoir computing, as depicted in Figure 6b, spanning from [0000] to [1111]. Leveraging the well-known efficiency of reservoir computing in complex tasks such as speech recognition, pattern recognition, and time series analysis, we employed our four-bit reservoir computing data from the TiN/WO_x/FTO memristor for

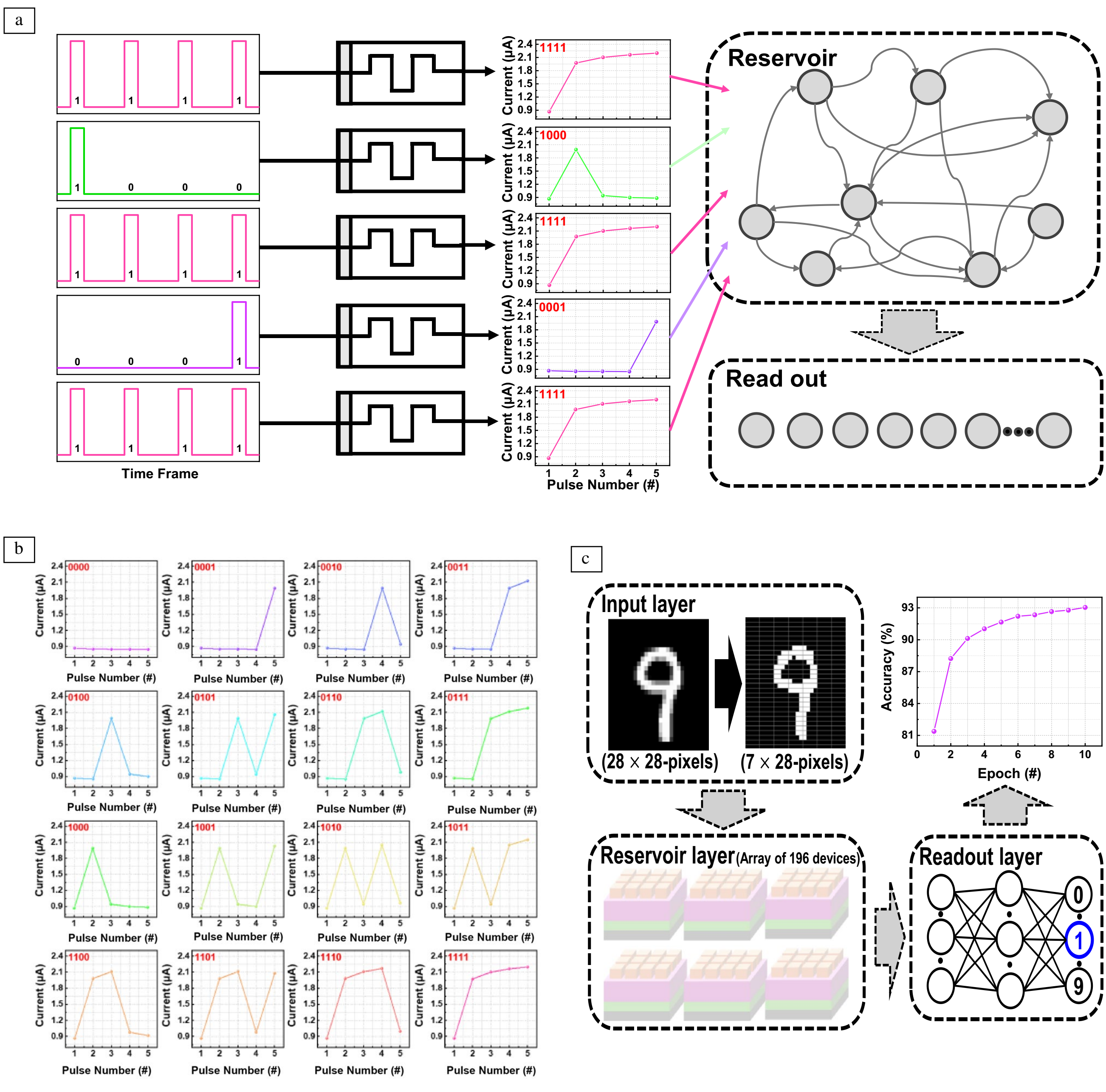

Figure 6. (a) Schematic illustration of reservoir computing implemented using TiN/WO_x/FTO memristor. (b) Implemented four-bit reservoir computing, ranging from [0000] to [1111]. (c) Pattern recognition system implemented through four-bit reservoir computing.

pattern recognition using artificial neural networks (ANNs). The system comprised an input layer, a reservoir layer, and a readout layer. The input layer processed modified images from the MNIST database, each originally 28 × 28 pixels, which were binarized into four-pixel patterns, resulting in four-pixel-patterned 7 × 28-pixel images (Figure 6c). Each of these four-pixel patterns corresponded to the states of the four-bit reservoir computing acquired from the TiN/WO_x/FTO memristor, with black pixels representing "0" state and white pixels representing "1" state. These binarized images were then mapped to physical reservoir states in the reservoir layer, comprising 196 arrays of memristors. Finally, through the readout layer, consisting of sequentially fully connected layers, the training and testing processes were conducted using 50,000 and 10,000 images, respectively. With 10 epochs, we achieved an accuracy exceeding 93%, demonstrating the commendable computing capabilities of our memristor, capable of facilitating versatile and high-density memory applications.

Conclusions

In this study, we successfully fabricated a synapse-emulating memristor comprised of TiN/WO_x/FTO stacks deposited on a glass substrate. Prior to synapse emulation, we conducted DC bias applications toward the top electrode to assess the electrical properties of the fabricated memristor. Our results demonstrate excellent uniformity across 10^2 repetitive cycles and 30 cycles measured in 12 different cells, showcasing a high average HRS/LRS memory window of 14.51 over 10^2 cycles, indicating outstanding properties. Moreover, we identified the volatile functions of the memristor through retention tests, observing rapid facilitation of current under pulse measurements. This facilitated the emulation of sensory memory utilizing the TiN/WO_x/FTO memristor, elucidated by the migration of oxygen ions within the insulating layer's-built gradient. Utilizing the large capacity and the lasting fading effect of sensory memory, we implemented a high-density 14×14 synapse image, along with PPF, brain-like learning functions, and reservoir computing, integrated into a pattern recognition system. These implementations showcased the memristor's favorable multifunctional behaviors. Furthermore, various synapse emulations were conducted through spike-dependent synaptic weight modulation techniques, including SRDP, SADP, and SWDP. Overall, the TiN/WO_x/FTO memristor investigated in this study holds significant promise for future neuromorphic computing applications, owing to its exceptional properties and versatile functionalities.

Author contributions

D.J.: Formal analysis, Investigation—performed the experiments, Resources, and Writing—original draft. M.N., G.K., and J.L.: Investigation—data/evidence collection, Computation, and Formal analysis. S.K.: Conceptualization, Writing—review & editing, Resources, Project administration, and Funding acquisition.

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Data availability

The data that support the findings of this study are available within the article and its supplementary material and from the corresponding authors upon reasonable request.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary information

The online version contains supplementary material available at https://doi.org/10.1557/s43577-024-00819-1.

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